Course Notes

Lecture 07 — ZigBee

Communication Technologies for the Internet of Things

Academic year 2026–2027

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Text to support independent study

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How to Use These Notes

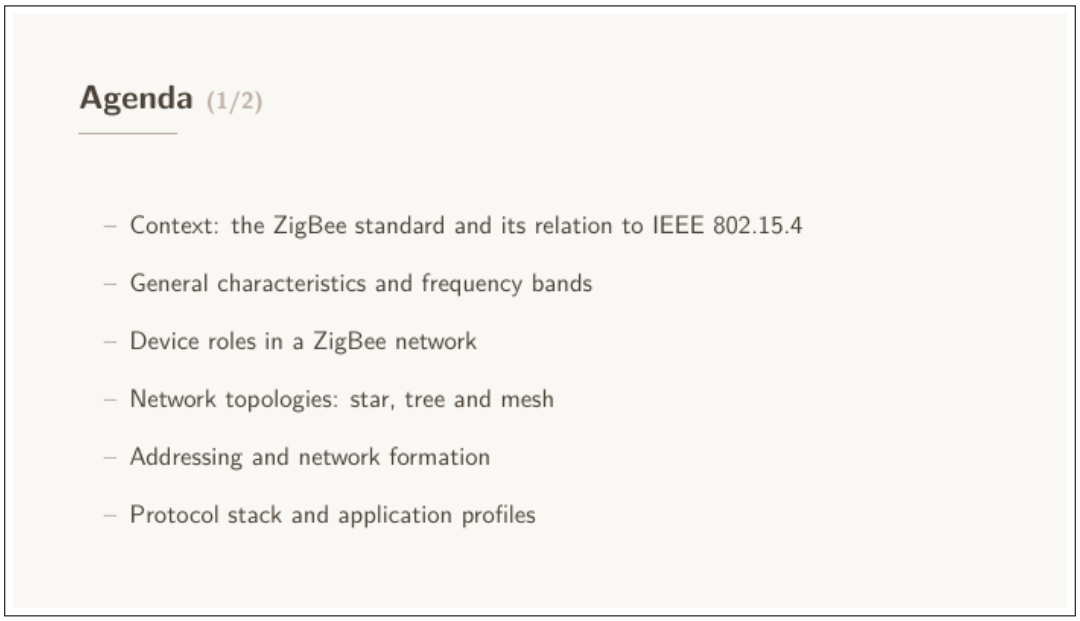
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

1. Commentary on the Slides

Slide 2. Agenda (1/2)



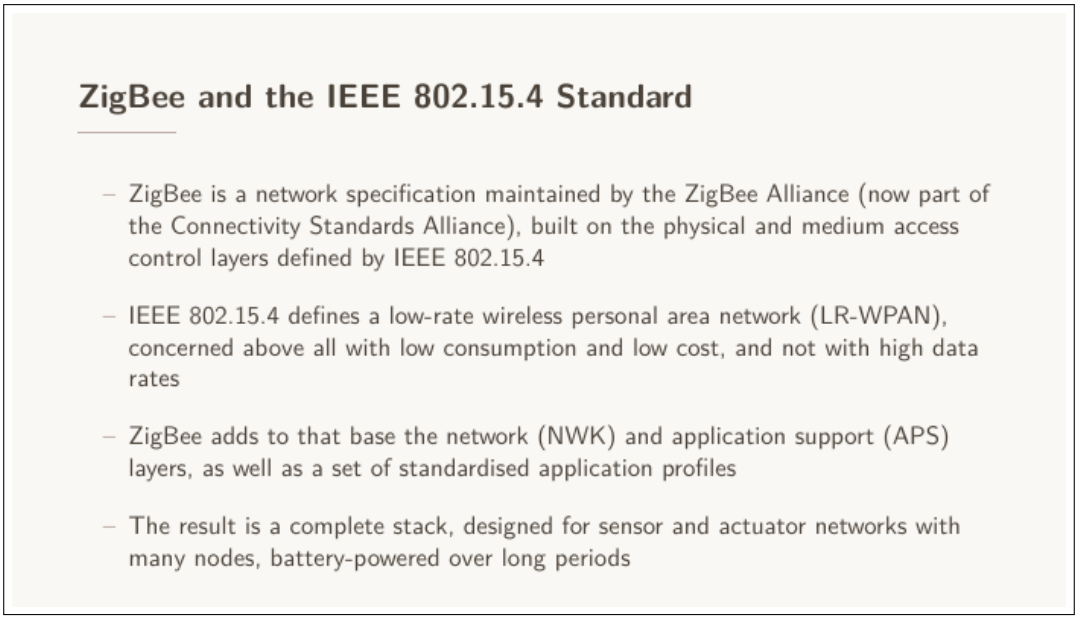
The path through the lecture. The lecture follows the construction of the technology from the bottom up: first the normative base, then the device roles, then the topologies those roles make possible, and finally the stack and profiles. This order has its own logic. ZigBee topologies are not an arbitrary design choice: they follow directly from what each type of device can or cannot do.

Slide 3. Agenda (2/2)



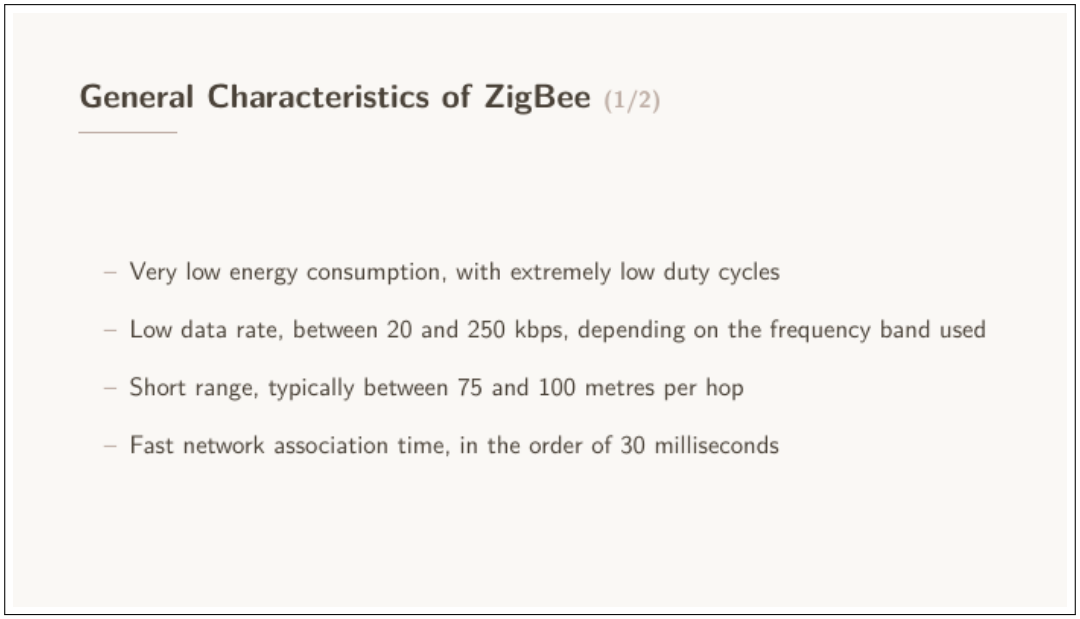
The two final topics. Energy consumption and use cases close the talk. They are best read together: the consumption profile is what determines where the technology applies.

Slide 4. ZigBee and the IEEE 802.15.4 Standard



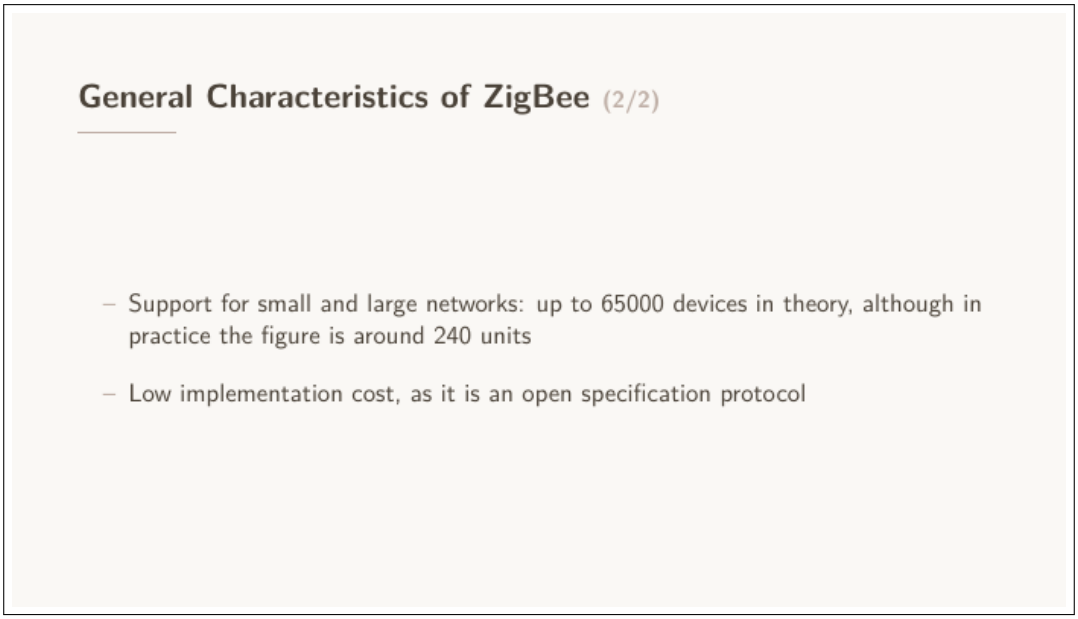
Two standards, one stack. ZigBee does not define a complete technology from scratch. It rests on the physical and medium access layers of IEEE 802.15.4, and adds the network and application support layers, plus a set of standardised profiles. This division of responsibilities is exam material. The IEEE defines how bits cross the air; the ZigBee Alliance defines how devices organise themselves and understand each other.

Slide 5. General Characteristics of ZigBee (1/2)



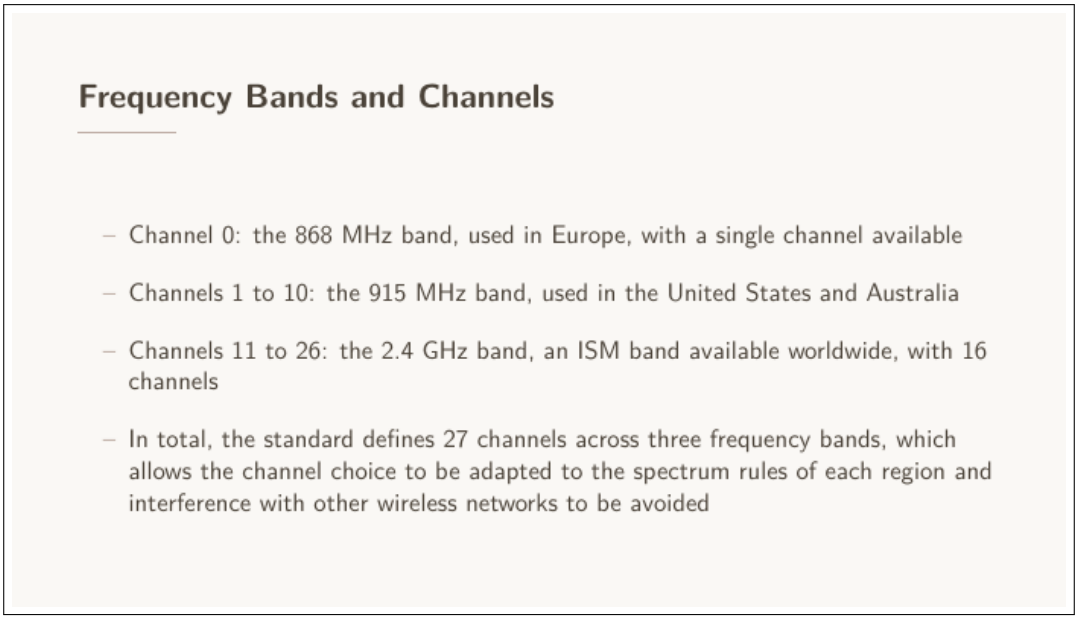
The numbers that characterise the technology. Very low consumption, a data rate between 20 and 250 kbps, a range of 75 to 100 metres per hop, and association to the network in about 30 milliseconds. The last value deserves a note. A fast association means the device does not need to stay active long to join the network, which in itself reduces consumption.

Slide 6. General Characteristics of ZigBee (2/2)



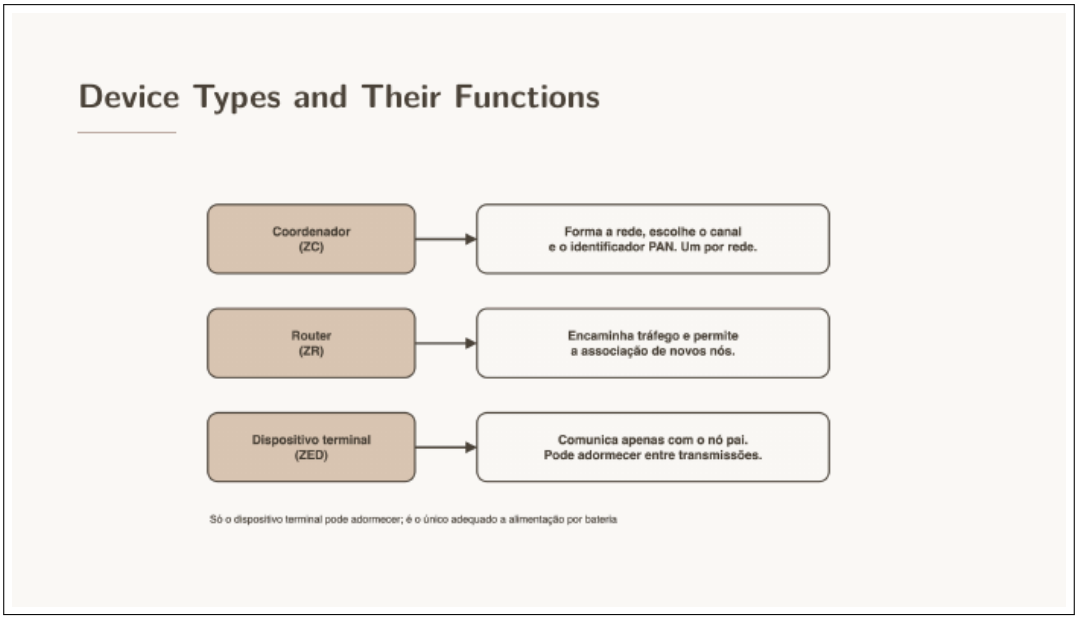
Theoretical scale and practical scale. The standard allows up to 65000 devices, but in practice the number is around 240 units. The discrepancy is not a defect: it results from memory limitations in intermediate nodes and from the management of routing tables. Knowing how to distinguish the limit of the standard from the limit of an implementation is an engineering skill.

Slide 7. Frequency Bands and Channels



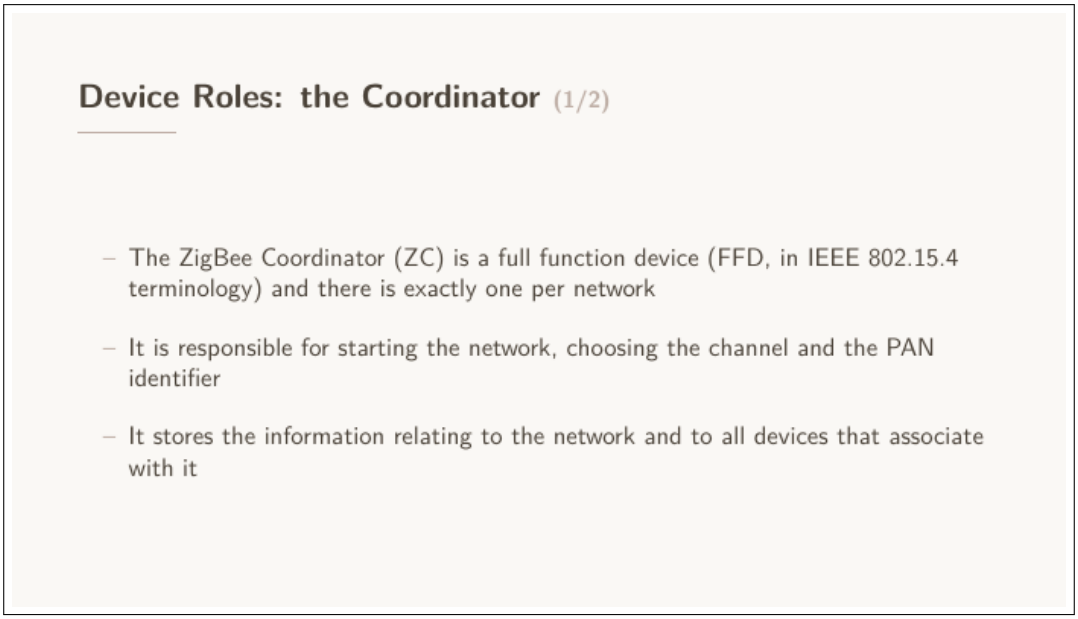
Where the technology operates. One channel in the 868 MHz band in Europe, ten in the 915 MHz band in the United States and Australia, and sixteen in the 2.4 GHz band, available worldwide. The existence of 27 channels lets the choice be adapted to the spectrum rules of each region and interference with other wireless networks, namely Wi-Fi, be avoided.

Slide 8. Device Types and Their Functions



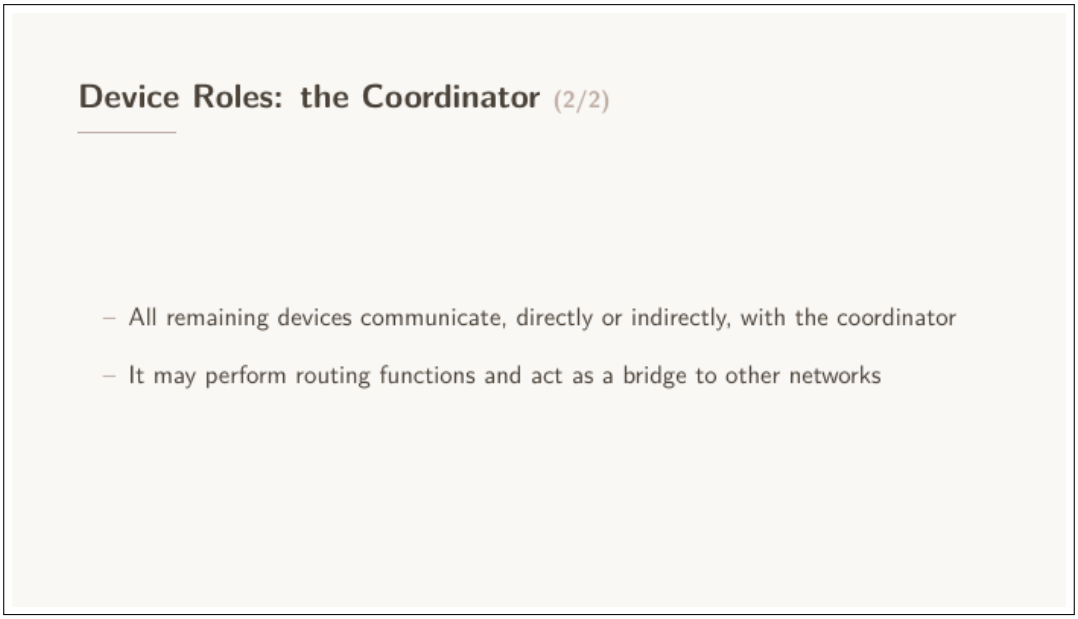
The three roles and what separates them. The diagram summarises the distinction that structures the whole lecture. The coordinator forms the network and there is one per network. The router forwards and allows associations. The end device communicates only with its parent. The footer sentence is the one to retain: only the end device can sleep, and it is therefore the only one suited to battery power.

Slide 9. Device Roles: the Coordinator (1/2)



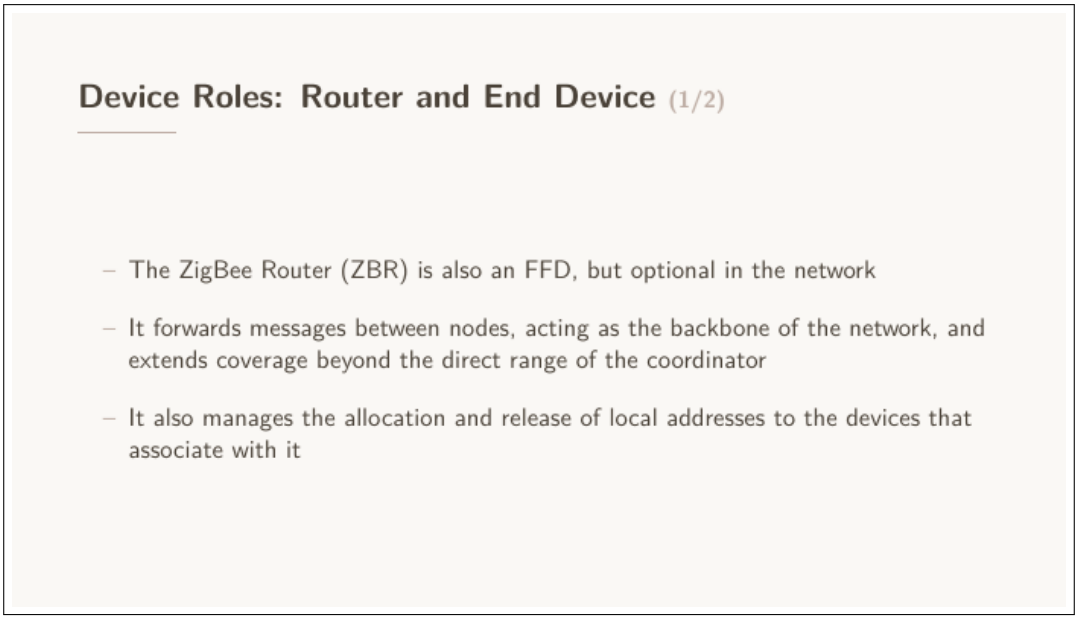
The coordinator. It is a full-function device and there is exactly one per network. It starts the network, choosing the channel and the PAN identifier, and keeps the information about all associated devices.

Slide 10. Device Roles: the Coordinator (2/2)



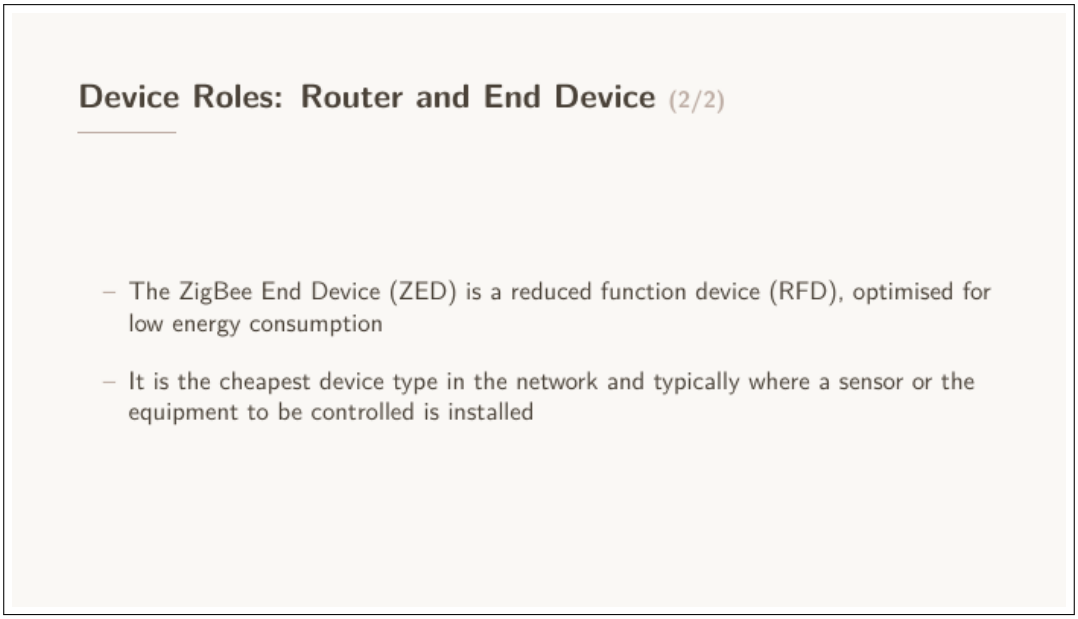
The coordinator's dual role. All the other devices communicate with it, directly or indirectly. It may also perform routing functions and act as a bridge to other networks. This last function anticipates the gateway concept already covered in Lecture 06.

Slide 11. Device Roles: Router and End Device (1/2)



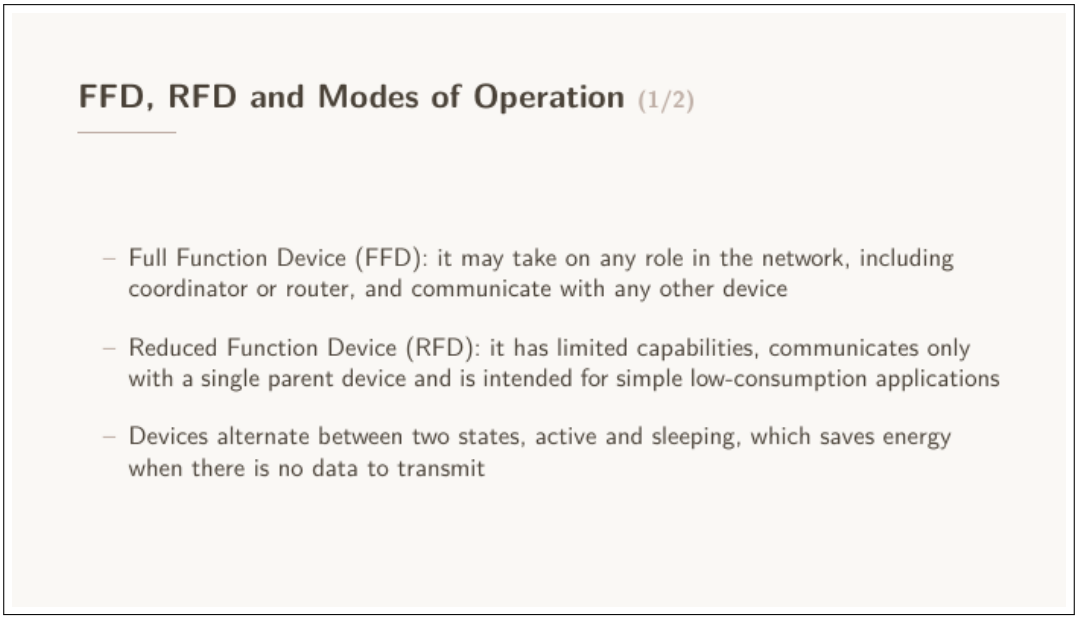
The router. Also a full-function device, but optional. It forwards messages, working as a backbone, and extends coverage beyond the coordinator's direct range. It also manages the assignment of local addresses to the devices that associate with it.

Slide 12. Device Roles: Router and End Device (2/2)



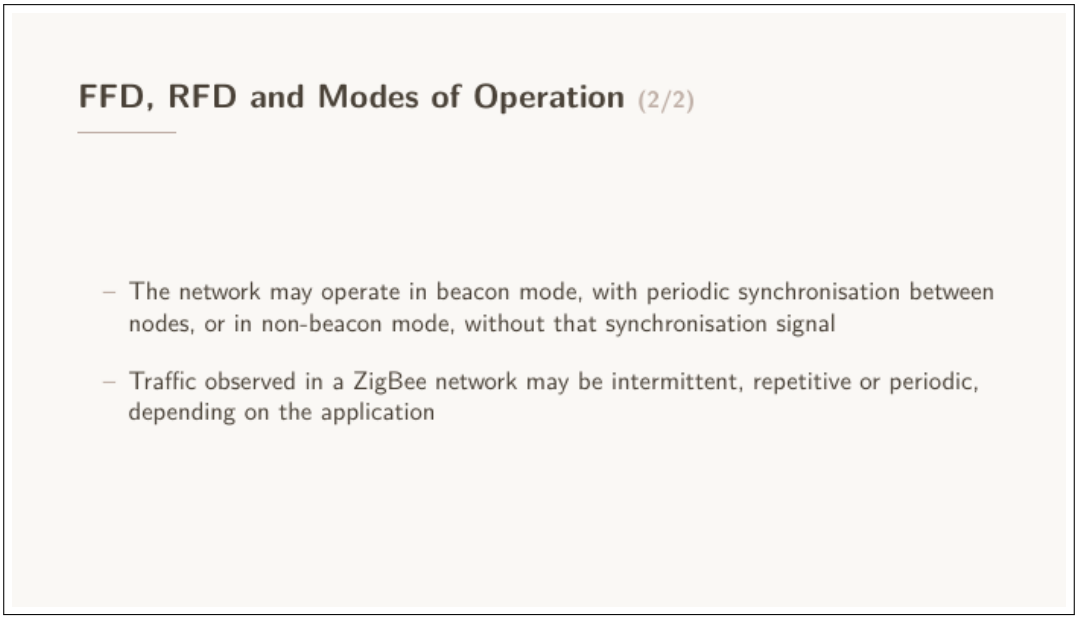
The end device. Reduced-function, optimised for low consumption. It is the cheapest type in the network and where the sensor or the equipment to be controlled is typically installed.

Slide 13. FFD, RFD and Modes of Operation (1/2)



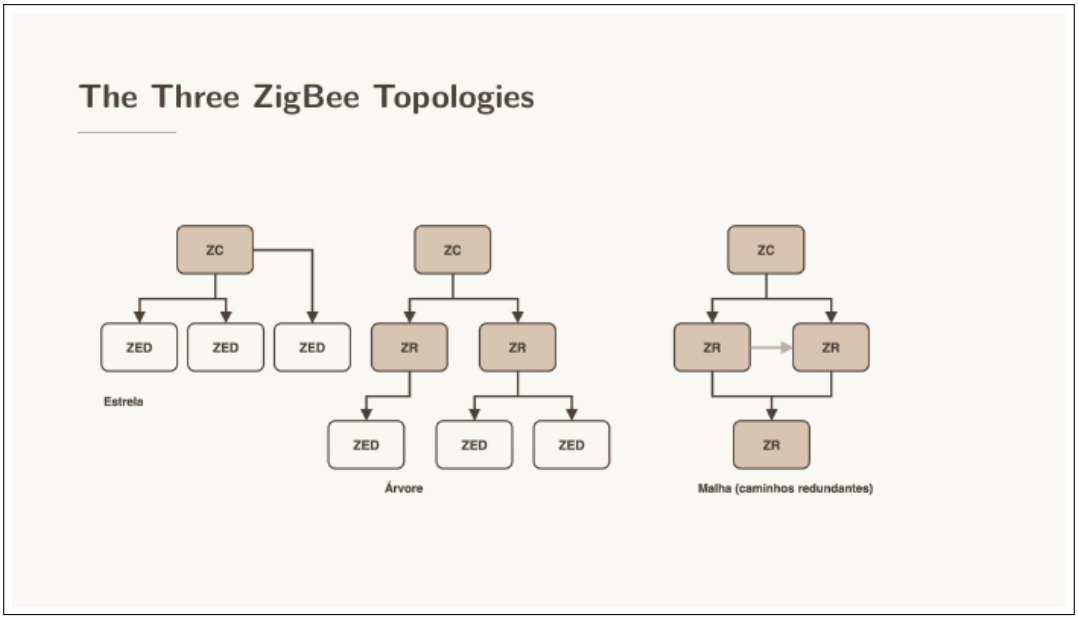
FFD and RFD. A full-function device can take any role and communicate with any other. A reduced-function one communicates only with a single parent. Devices alternate between active and sleeping, which saves energy when there is no data to transmit.

Slide 14. FFD, RFD and Modes of Operation (2/2)



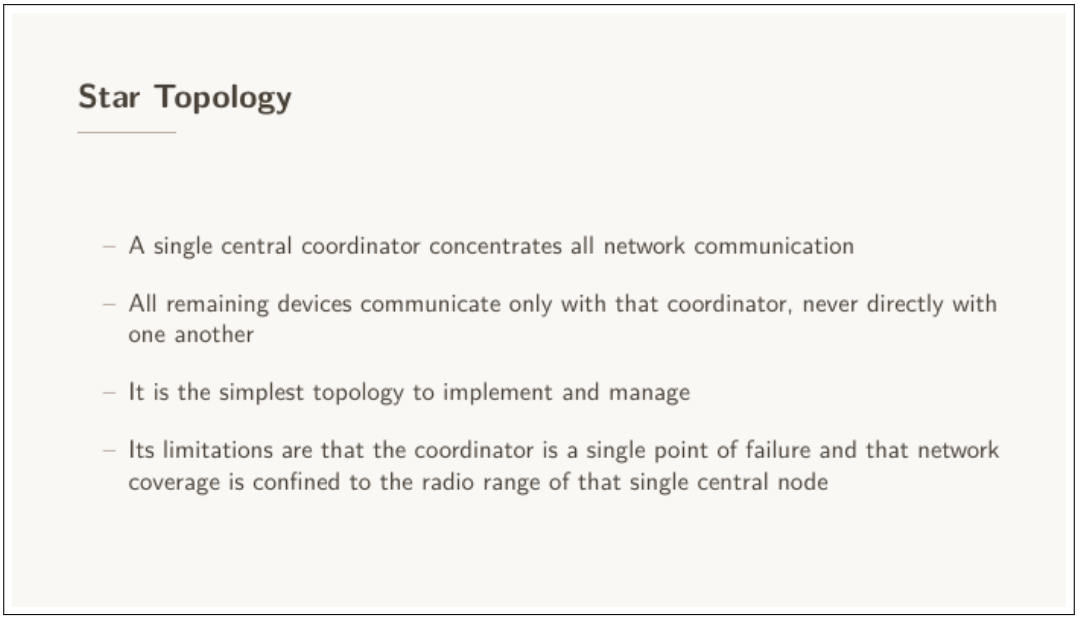
Beacon mode and traffic types. The network may operate with periodic synchronisation by beacons, or without it. Traffic may be intermittent, repetitive or periodic, depending on the application. The choice of mode has a direct consequence on the admissible topology, as will be seen on slide 17.

Slide 15. The Three ZigBee Topologies



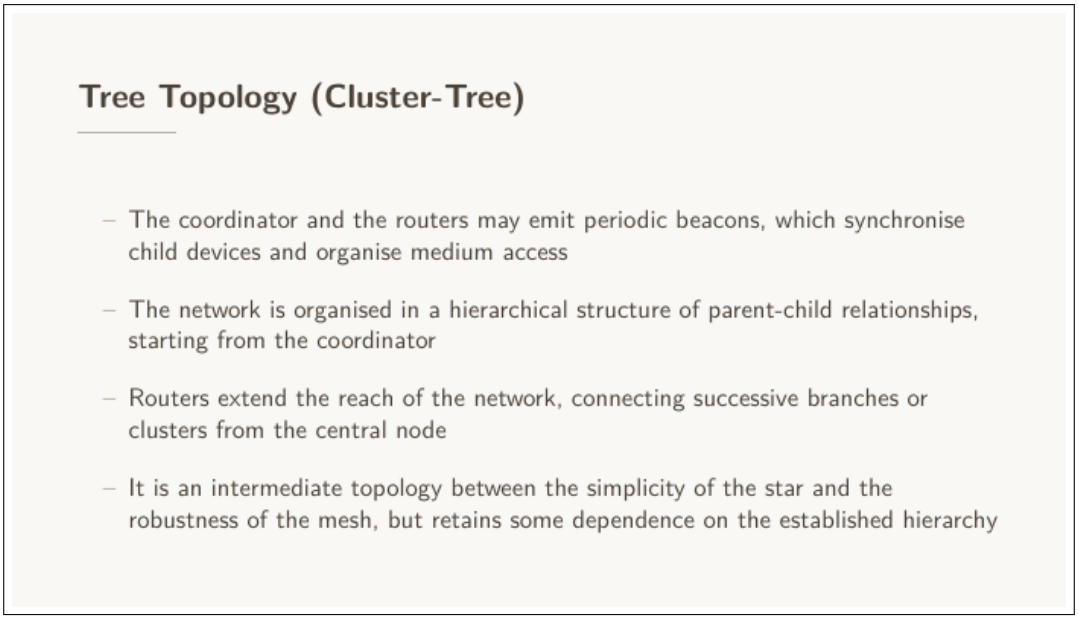
The three topologies, side by side. The diagram allows a comparison at a glance. In the star, everyone connects to the coordinator. In the tree, routers create branches. In the mesh, the dashed links between routers create redundant paths. It is those dashed links that make all the difference, and it is on them that the lab session focuses.

Slide 16. Star Topology



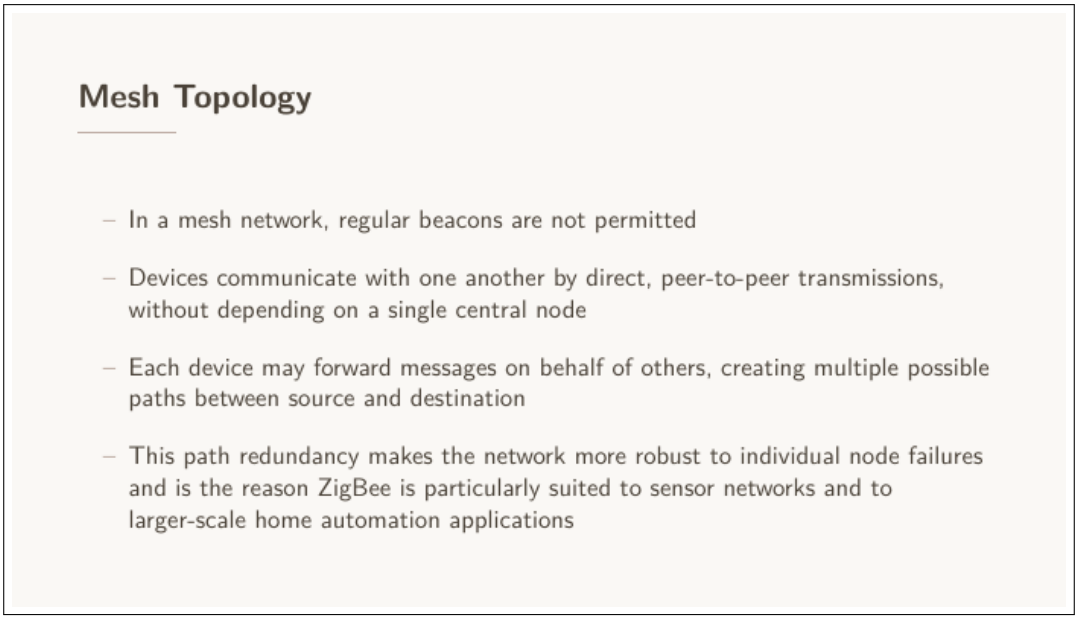
The star. A central coordinator concentrates all communication, and the other devices never communicate directly with each other. It is the simplest topology to implement and manage. It has two limitations: the coordinator is a single point of failure, and coverage is confined to that node's radio range.

Slide 17. Tree Topology (Cluster-Tree)



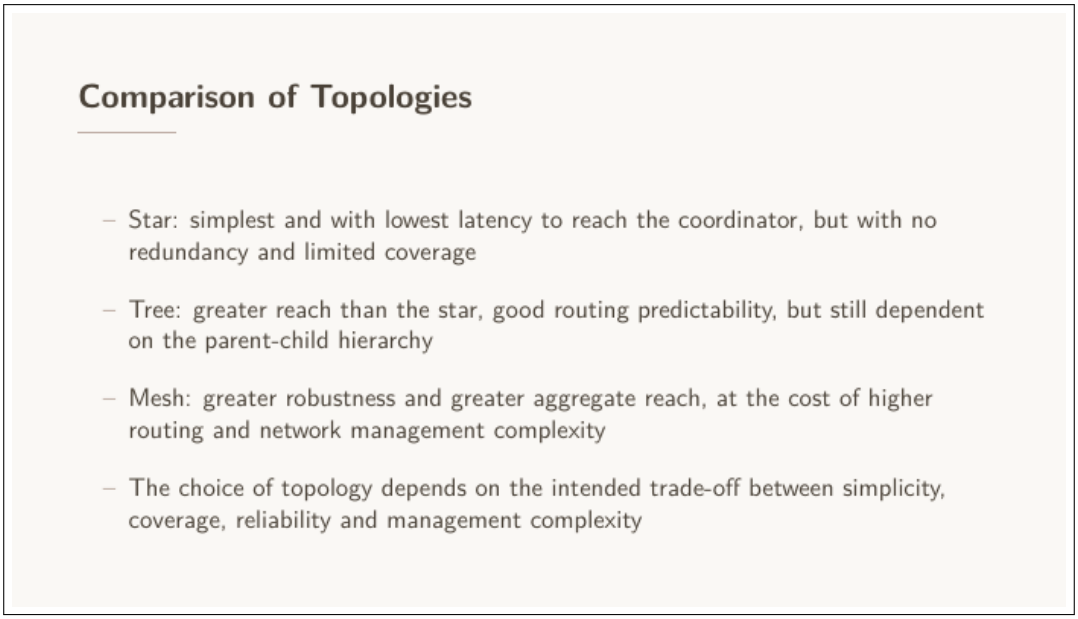
The tree. The coordinator and routers emit periodic beacons, which synchronise the children and organise medium access. The network is organised in parent and child relationships. Routers extend range by linking successive branches. It is an intermediate topology, but it keeps a dependence on the hierarchy established.

Slide 18. Mesh Topology



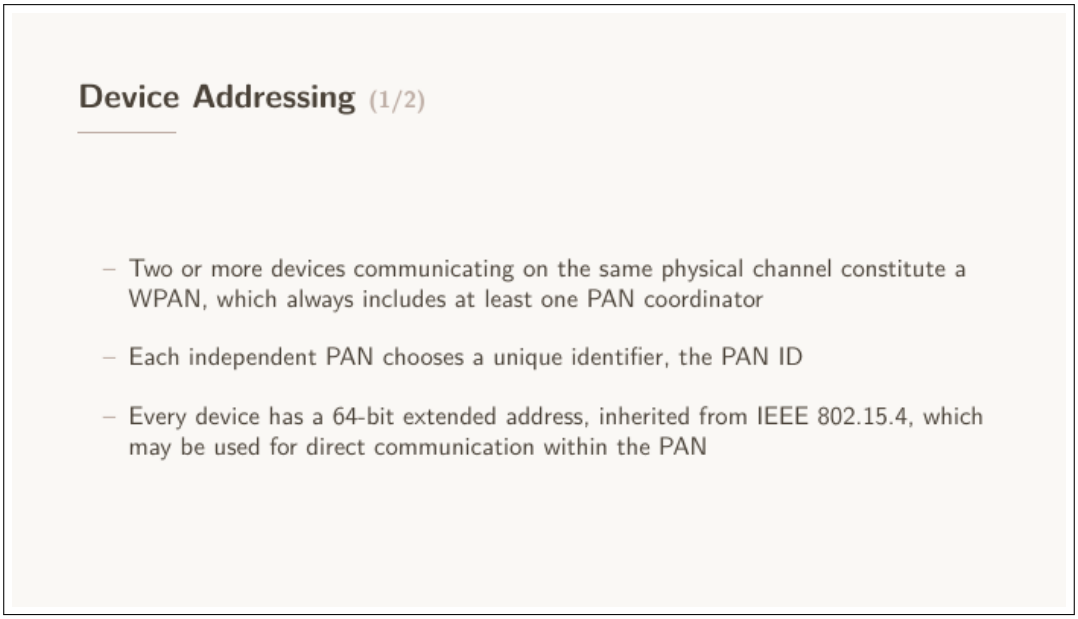
The mesh. In a mesh network regular beacons are not allowed. Devices communicate by direct transmissions, and each may forward on behalf of others. This redundancy of paths results in robustness to failures of individual nodes. It is the reason why ZigBee is suited to sensor networks and larger-scale home automation.

Slide 19. Comparison of Topologies



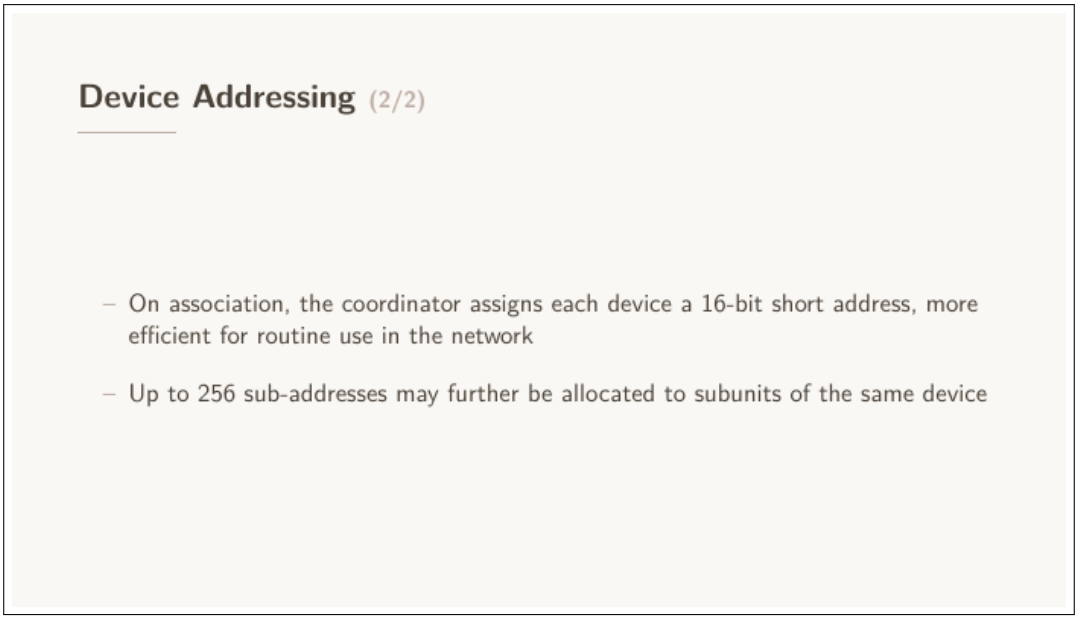
The comparison table. The star is simpler and has lower latency to the coordinator, but has no redundancy. The tree has greater range and predictable routing. The mesh has greater robustness and aggregate range, at the cost of greater management complexity. The conclusion to retain is the last one: the choice depends on the compromise sought, and there is no best topology in absolute terms.

Slide 20. Device Addressing (1/2)



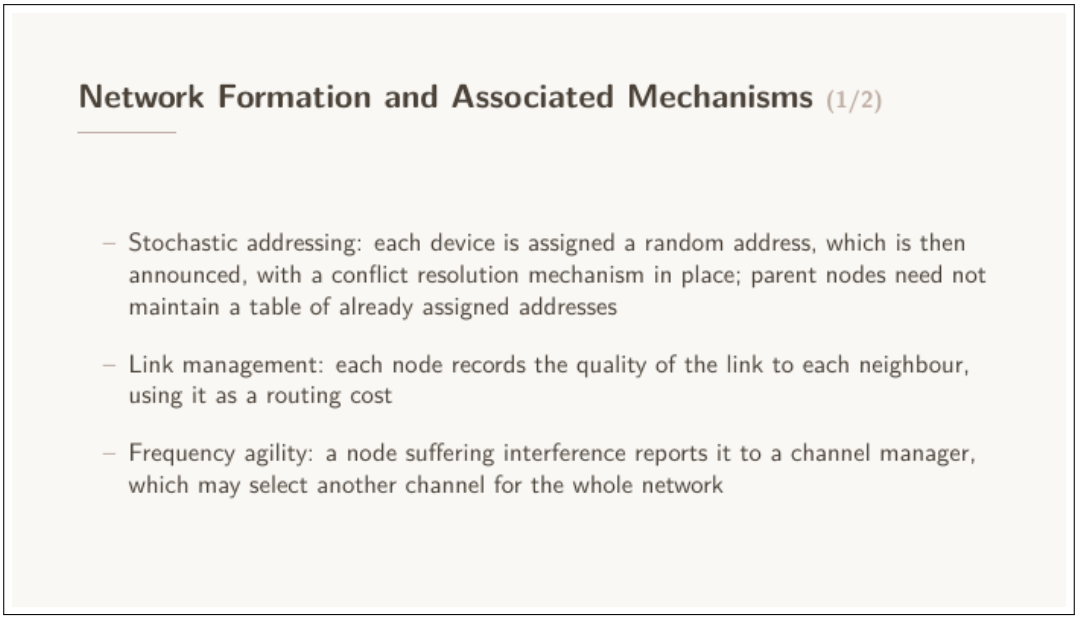
Network and device identifiers. Two or more devices on the same channel form a WPAN, which always includes a coordinator. Each independent PAN chooses a unique identifier. Every device has a 64-bit extended address, inherited from IEEE 802.15.4.

Slide 21. Device Addressing (2/2)



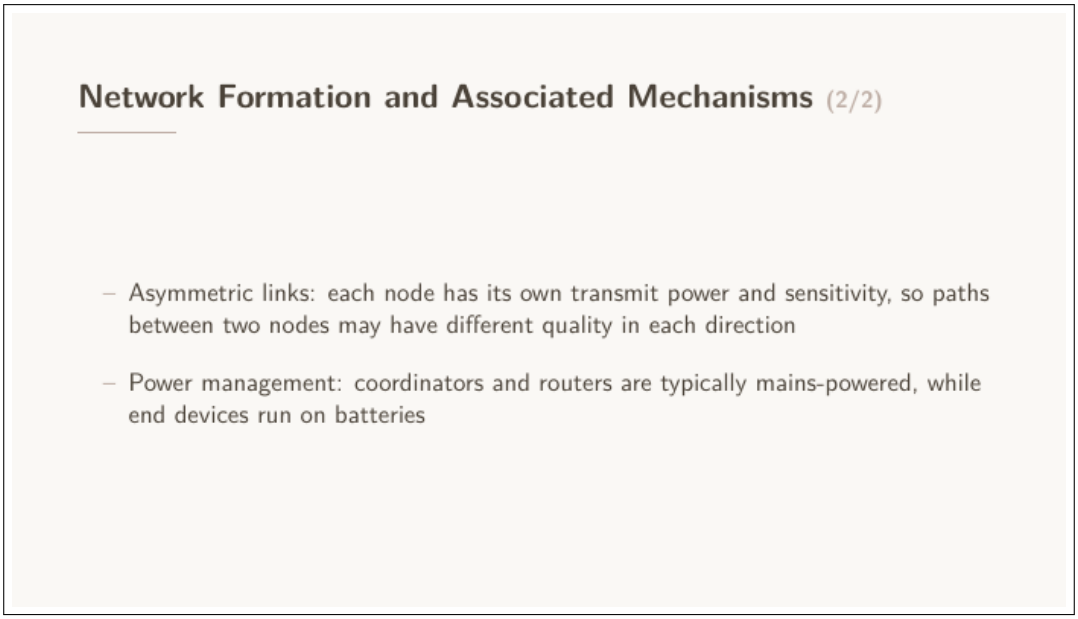
Short addresses and sub-addresses. On association, the coordinator assigns a 16-bit short address, more efficient for everyday use. Up to 256 sub-addresses may also be allocated to sub-units of the same device. Having two addresses per device, a long permanent one and a short assigned one, is the same pattern found in other technologies and is worth recognising.

Slide 22. Network Formation and Associated Mechanisms (1/2)



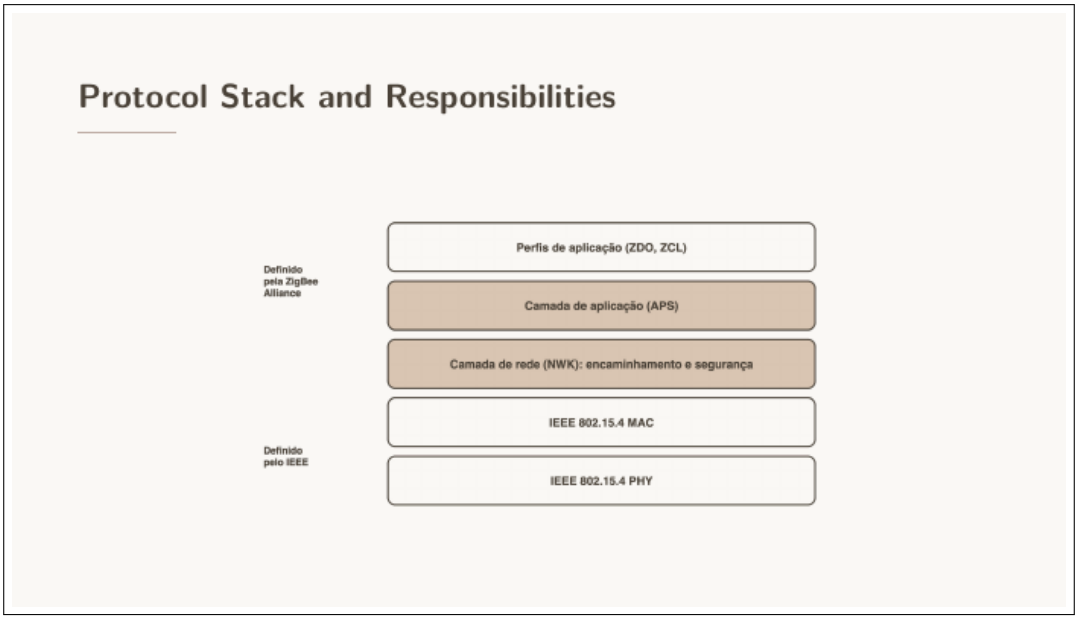
Four mechanisms of network formation. Stochastic addressing assigns random addresses with conflict resolution, dispensing with tables in parent nodes. Link management records the quality of each neighbour and uses it as a routing cost. Frequency agility lets a node suffering interference report to a channel manager, which may change the channel of the whole network.

Slide 23. Network Formation and Associated Mechanisms (2/2)



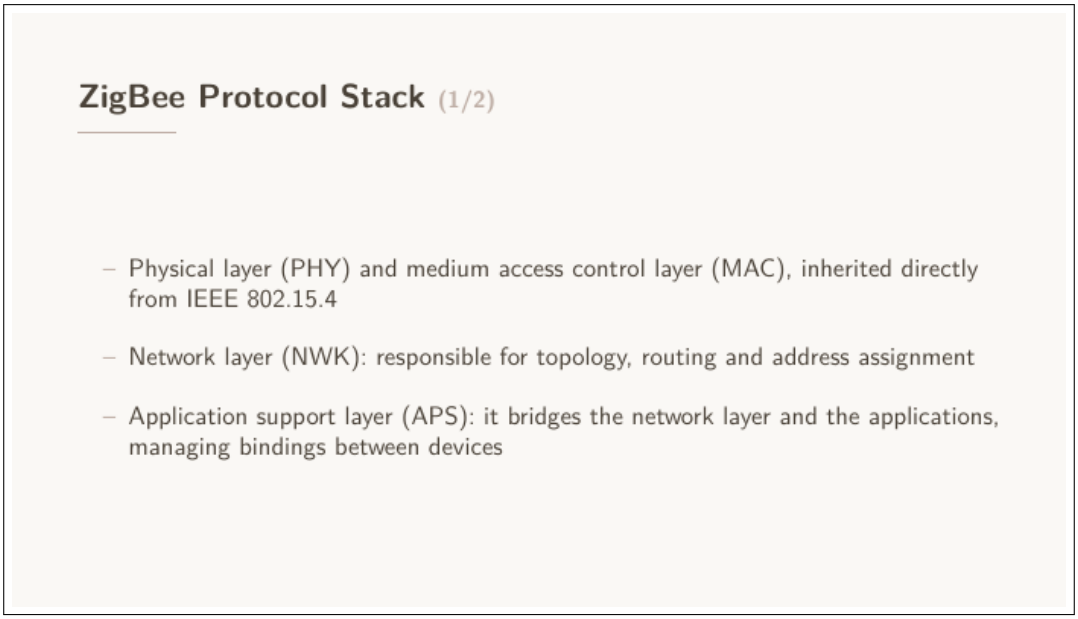
Asymmetry and power. Each node has its own power and sensitivity, so a path may have different quality in each direction. Coordinators and routers are typically mains-powered; end devices run on battery. This last observation closes the argument begun on slide 8.

Slide 24. Protocol Stack and Responsibilities



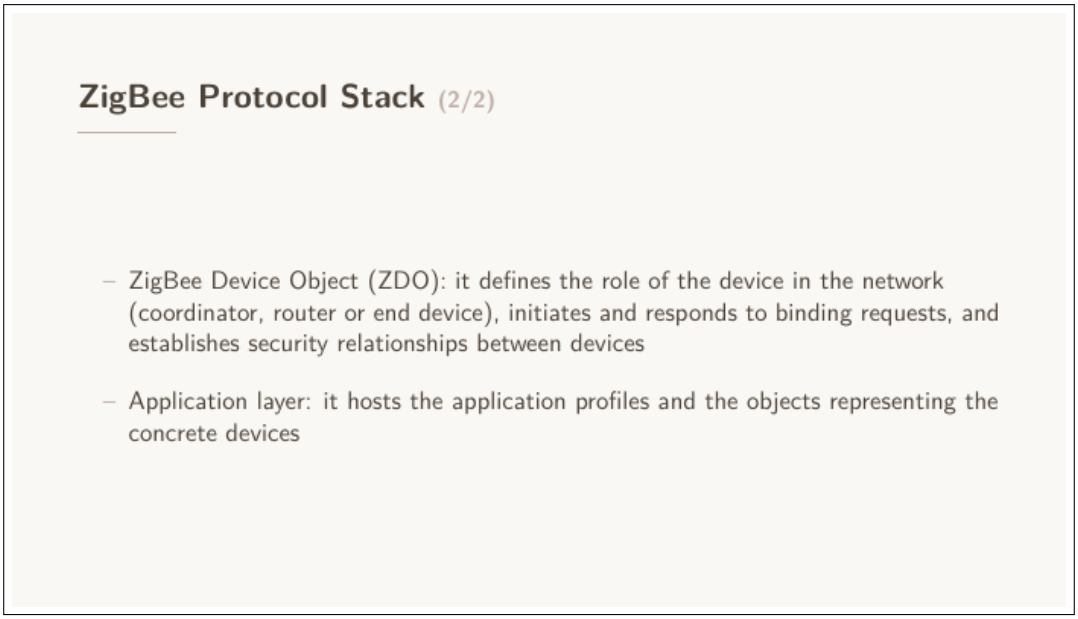
Who defines what. The diagram visually separates the layers defined by the IEEE from those defined by the ZigBee Alliance. Two layers below, three above. It is the fastest way to review the division of responsibilities introduced on slide 4.

Slide 25. ZigBee Protocol Stack (1/2)



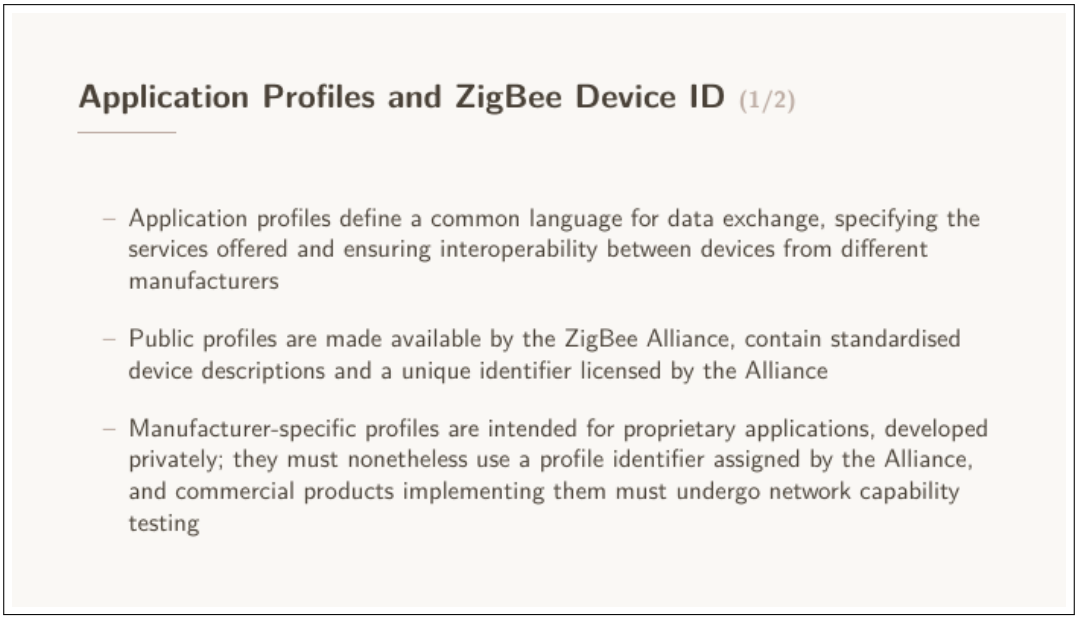
The network and application layers. The physical and medium access layers come from IEEE 802.15.4. The network layer handles topology, routing and address assignment. The application support layer bridges to applications and manages the associations between devices.

Slide 26. ZigBee Protocol Stack (2/2)



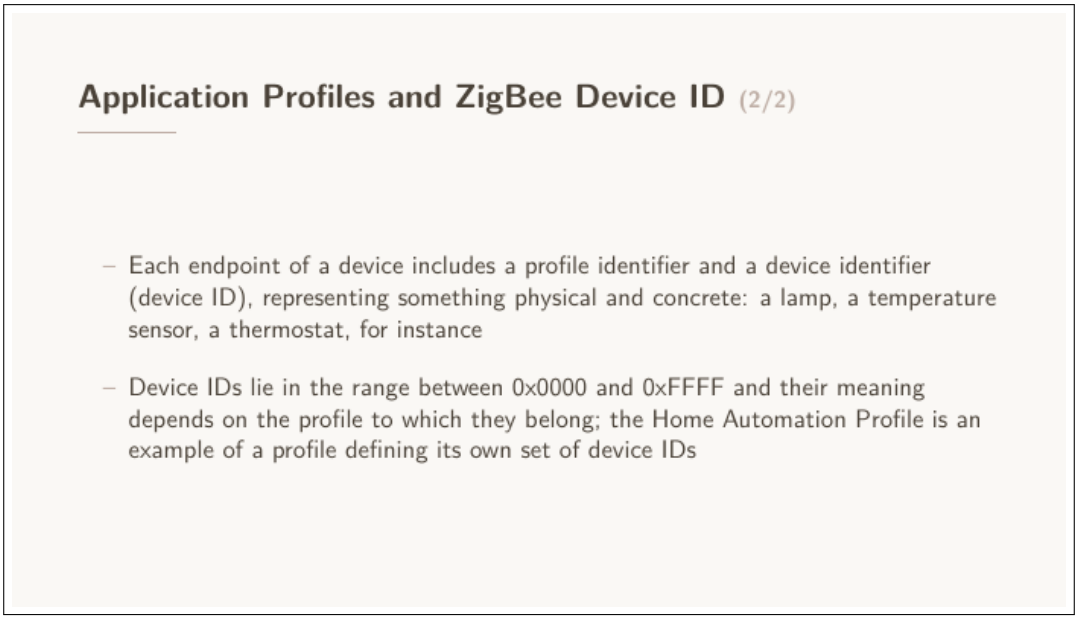
The ZigBee Device Object. The ZDO defines the device's role in the network, initiates and answers association requests, and establishes security relationships. Above it, the application layer hosts the profiles and the objects that represent concrete devices.

Slide 27. Application Profiles and ZigBee Device ID (1/2)



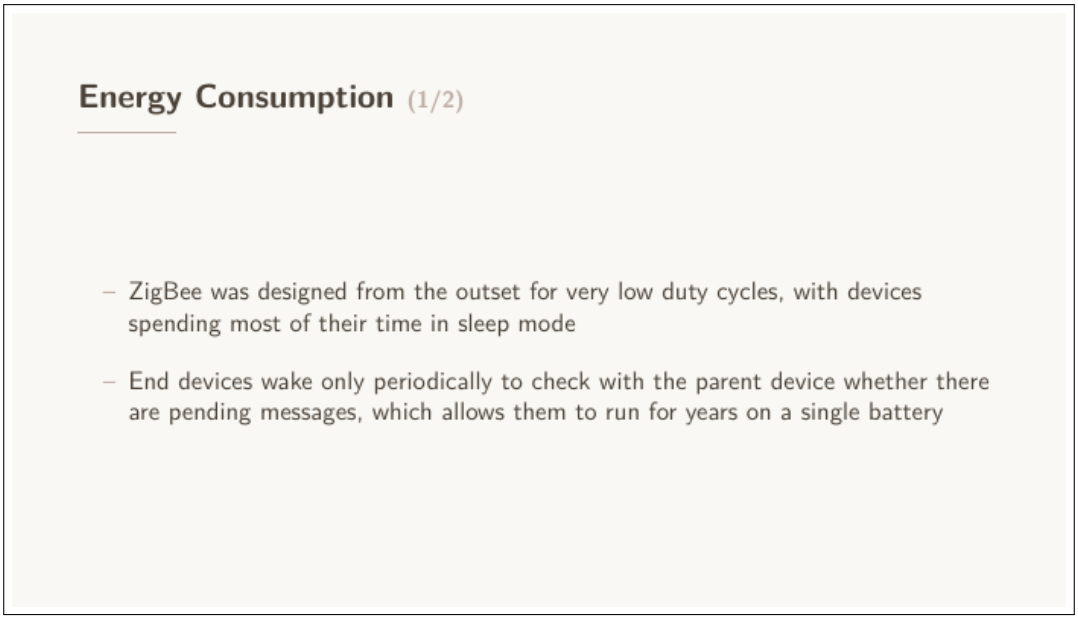
What profiles are for. Profiles define a common language for data exchange, specifying the services offered and guaranteeing interoperability between manufacturers. Public profiles come from the Alliance. Manufacturer-specific ones are for proprietary applications, but still have to use an identifier assigned by the Alliance.

Slide 28. Application Profiles and ZigBee Device ID (2/2)



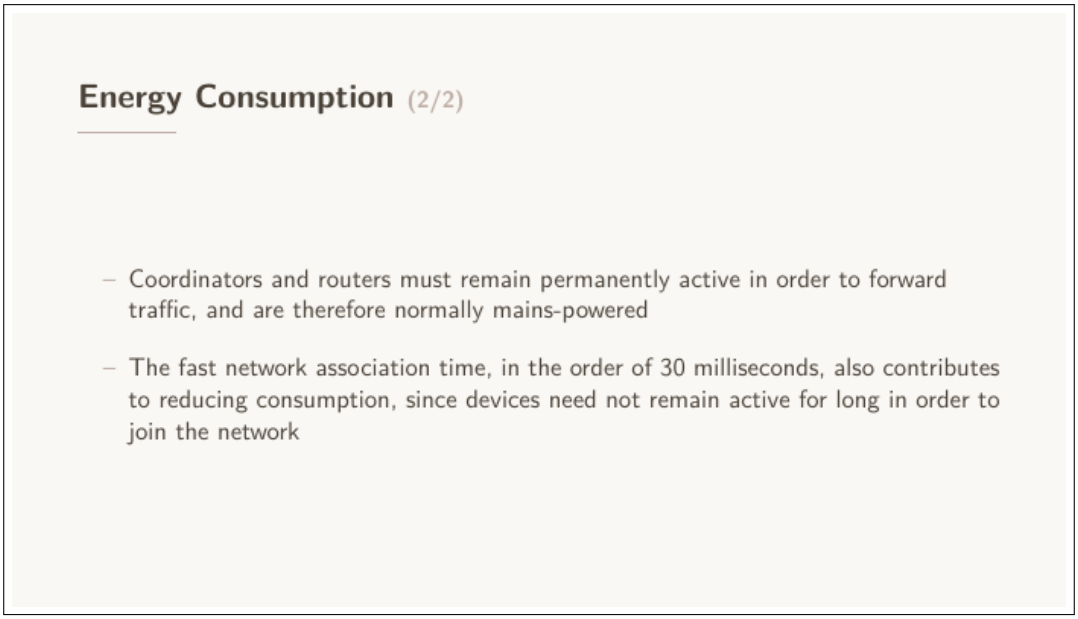
The device identifier. Each endpoint includes a profile identifier and a device identifier, the latter representing something physical and concrete: a lamp, a sensor, a thermostat. The meaning of the identifier depends on the profile it belongs to. The same numeric value designates different things in different profiles, which is a common source of confusion.

Slide 29. Energy Consumption (1/2)



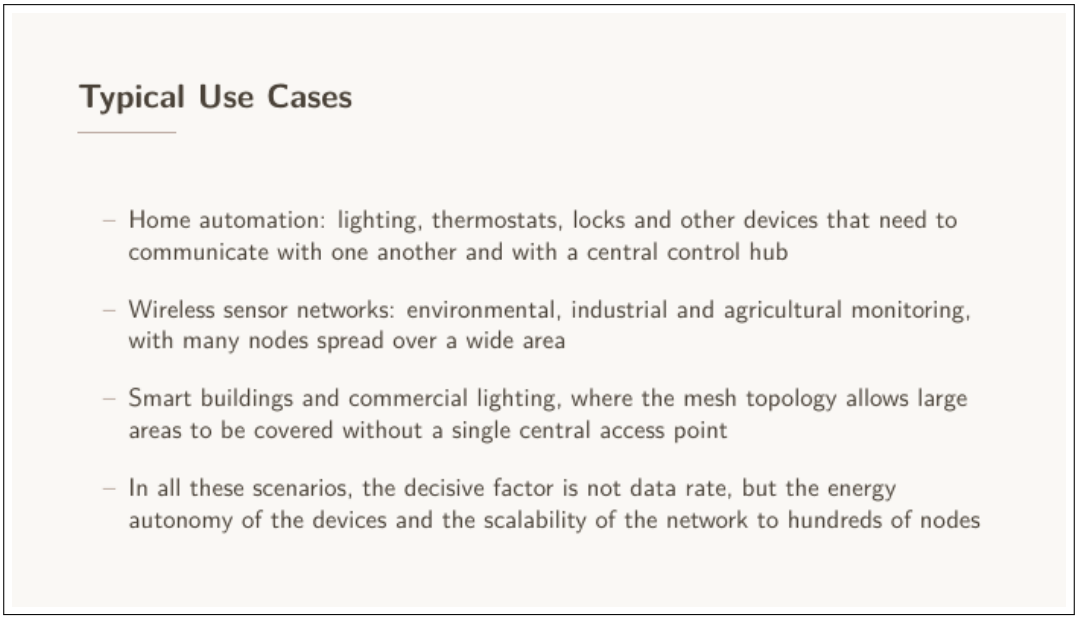
Where autonomy comes from. The technology was designed for very low duty cycles, with devices sleeping most of the time. End devices wake periodically only to check with the parent whether there are pending messages, which lets them run for years on a single battery.

Slide 30. Energy Consumption (2/2)



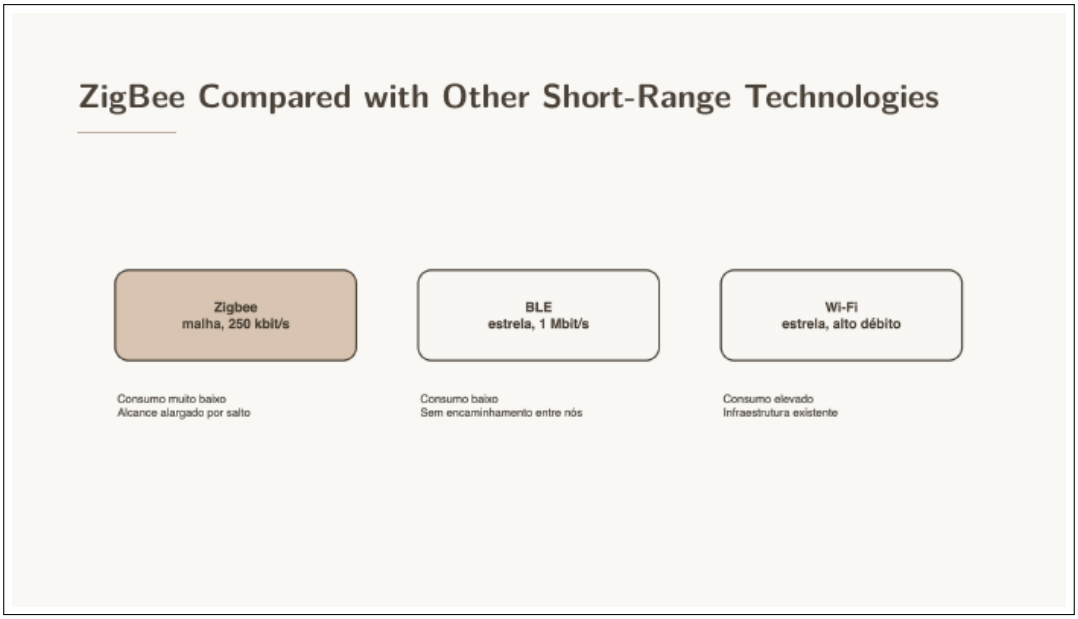
The price of forwarding. Coordinators and routers have to be permanently active in order to forward, and are therefore mains-powered. This asymmetry of power is an installation constraint that should be considered in design: routers have to be where there is a socket.

Slide 31. Typical Use Cases



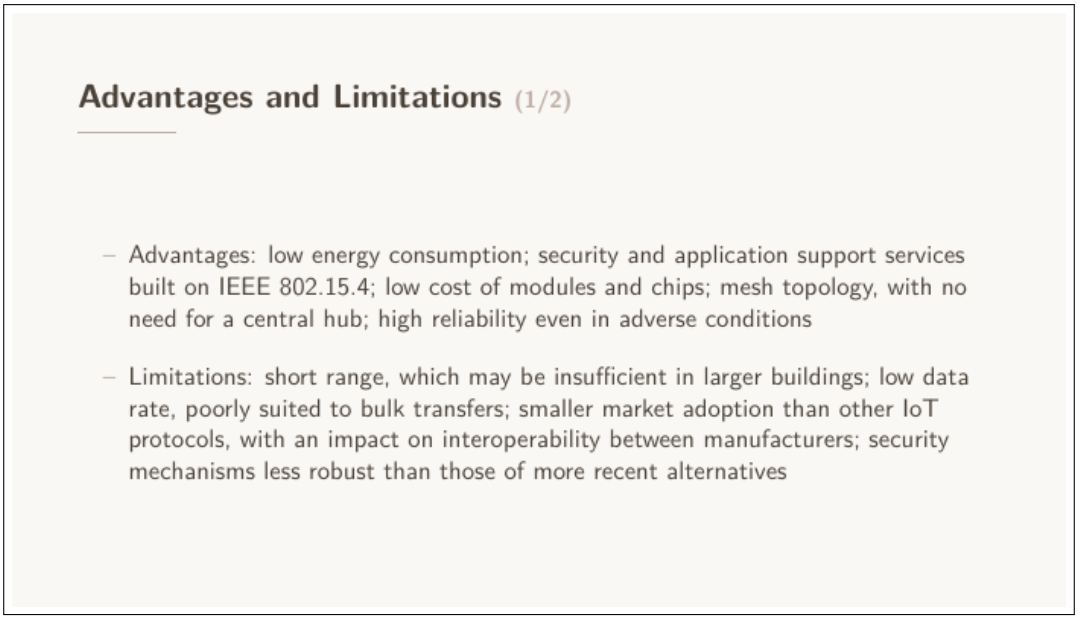
Where the technology is used. Home automation, wireless sensor networks and smart buildings. In all these scenarios the decisive factor is not data rate, but energy autonomy and scalability to hundreds of nodes.

Slide 32. ZigBee Compared with Other Short-Range Technologies



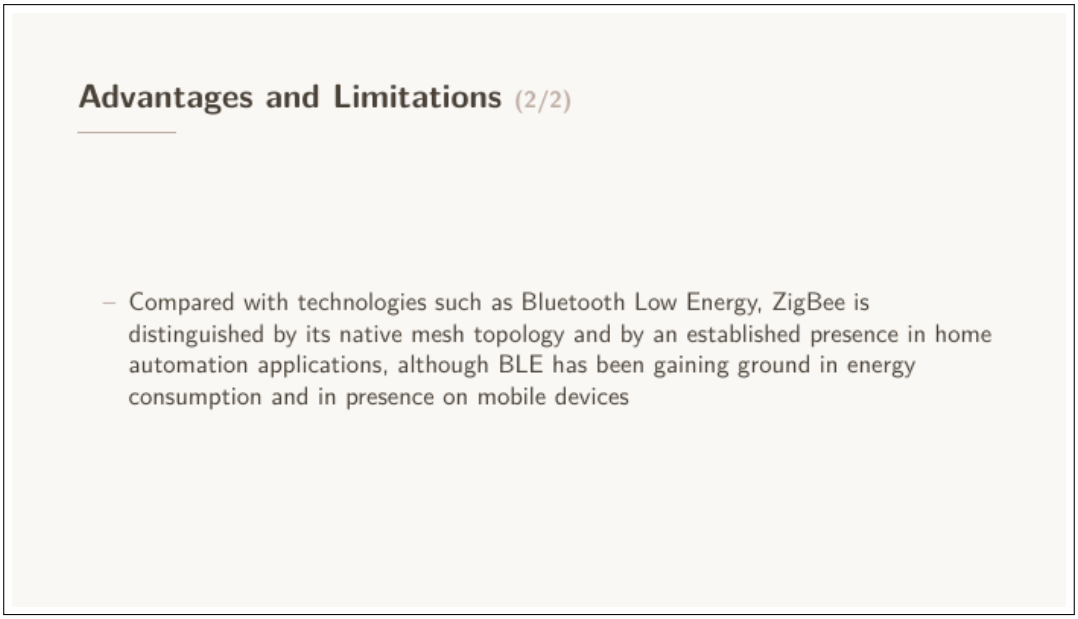
ZigBee against BLE and Wi-Fi. The diagram compares the three technologies in data rate, topology and consumption. ZigBee stands out for native mesh, which gives it extended range per hop. It is worth placing next to this diagram the one from Lecture 04 on BLE topologies, and checking that the structural difference between the two technologies lies precisely there.

Slide 33. Advantages and Limitations (1/2)



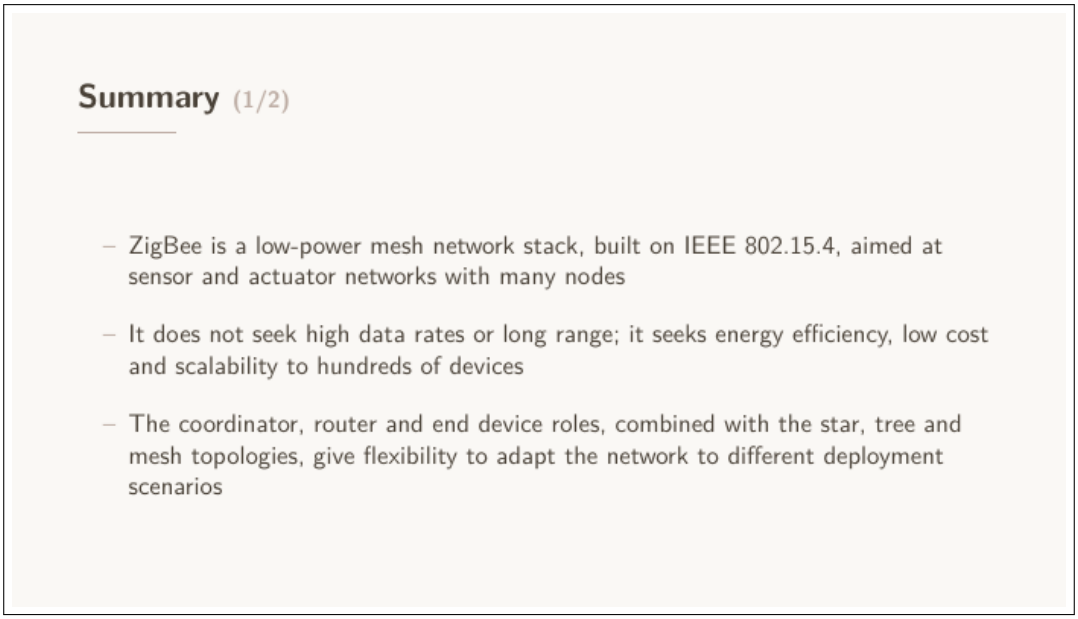
The balance. The advantages are low consumption, low cost, mesh with no central hub and reliability. The limitations are short range, low data rate, smaller market adoption and security mechanisms less robust than recent alternatives. A technology is not assessed by a list of virtues, it is assessed by its fit to the problem.

Slide 34. Advantages and Limitations (2/2)



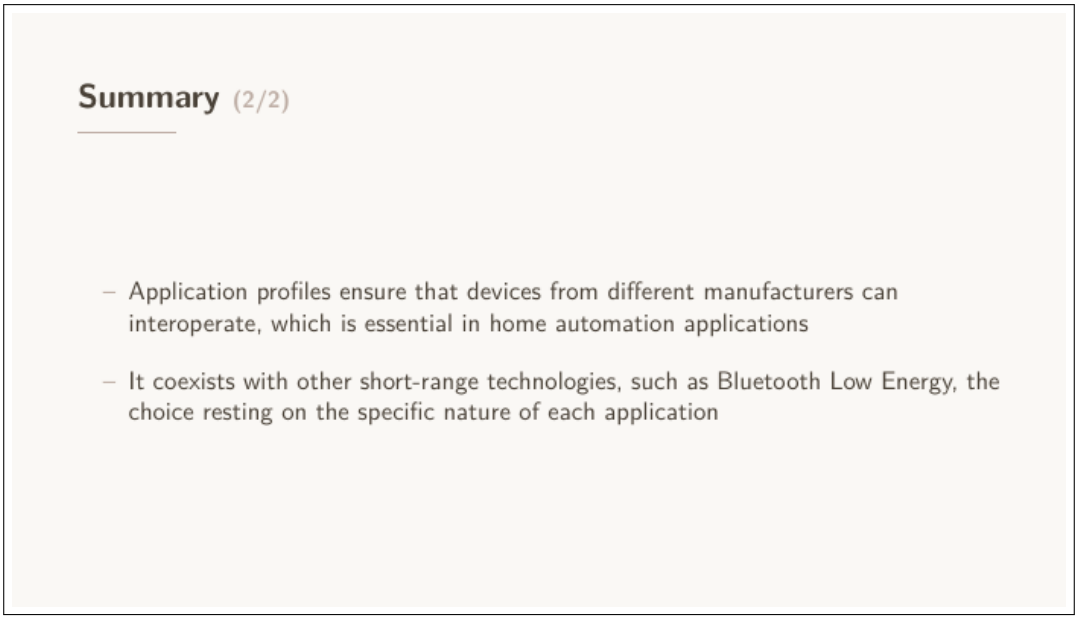
Comparison with BLE. Compared with Bluetooth Low Energy, ZigBee stands out for native mesh and a consolidated presence in home automation, although BLE has gained ground in consumption and in presence in mobile devices.

Slide 35. Summary (1/2)



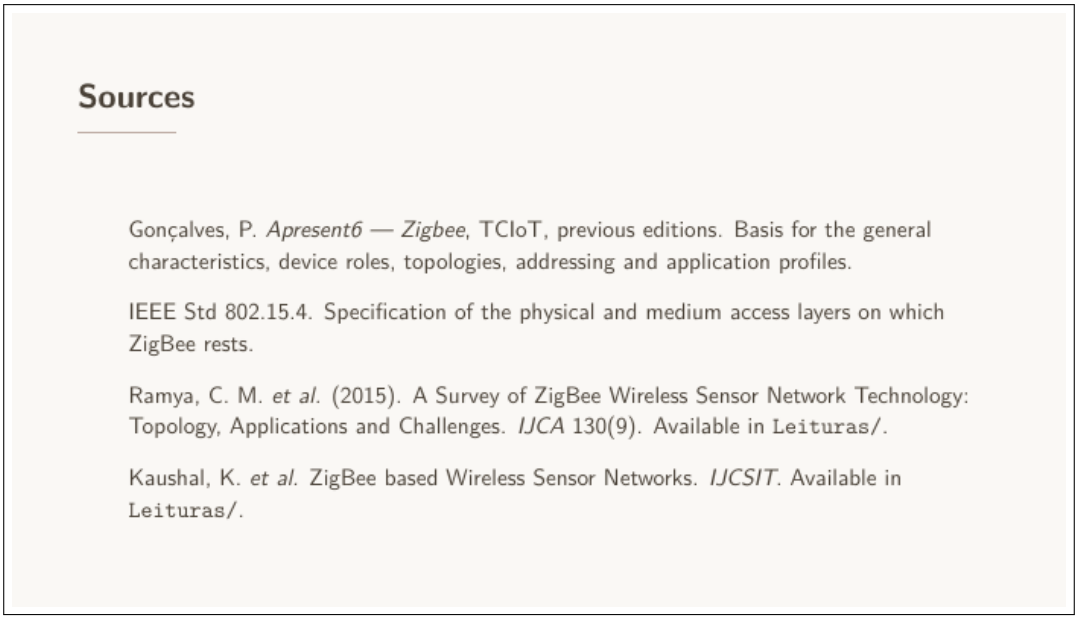
Summary, first part. ZigBee is a low-power mesh network stack, built on IEEE 802.15.4. It does not seek high data rates or ranges, it seeks energy efficiency, low cost and scalability. The three roles, combined with the three topologies, give flexibility to adapt the network to different scenarios.

Slide 36. Summary (2/2)



Summary, second part. Application profiles guarantee that devices from different manufacturers interoperate, which is essential in home automation. The technology coexists with BLE, and the choice depends on the concrete nature of each application.

Slide 37. Sources



Origin of the material. The basis is the course material from previous editions, complemented by the IEEE 802.15.4 standard and by two review papers available in the readings folder.

Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. State which layers are defined by IEEE 802.15.4 and which by the ZigBee Alliance.
2. Explain why only the end device can be battery-powered.
3. Describe what happens to the network when a router fails, in tree and in mesh topology.
4. Justify the discrepancy between the 65000 devices of the standard and the 240 units in practice.
5. Explain why there are two addresses per device, one of 64 bits and one of 16 bits.
6. State what frequency agility is and what problem it solves.
7. Compare the ZigBee topology with the BLE topology studied in Lecture 04, identifying the structural difference.

References

- Gonçalves, P. *Apresent6 — Zigbee*, TCIoT, previous editions.
- IEEE Std 802.15.4. Specification of the physical and medium access layers.
- Ramya, C. M. *et al.* (2015). A Survey of ZigBee Wireless Sensor Network Technology: Topology, Applications and Challenges. *IJCA* 130(9).
- Kaushal, K. *et al.* ZigBee based Wireless Sensor Networks. *IJCSIT*.